

Secretary of State Audit Report

Kate Brown, Secretary of State

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Health and Human Service Caseload Forecasting: Ways to Increase Confidence

Summary

The objective of our audit was to determine if adequate independence, oversight, and transparency exists for the Office of Forecasting, Research, and Analysis (forecast unit) to ensure objective and reasonable caseload and expenditure forecasts. The forecast unit is a shared service of the Department of Human Services (DHS) and the Oregon Health Authority (OHA). The unit's caseload forecasts are used to estimate expenditures for the two agencies' budgets.

Accurate caseload forecasts are important for budgeting. Although there is inherent uncertainty in forecasting caseloads, objective and impartial forecasts are important for developing accurate program budget estimates. If estimated expenditures are overstated, the legislature may initially approve a larger budget than necessary to meet actual demand. Yet, an under-estimated forecast may require a new appropriation from state General Fund reserves.

Agencies may have an incentive to create a cushion in their budgets by over-estimating caseload forecasts. A budget surplus allows an agency to cover budget shortfalls with an Emergency Board rebalance instead of requesting new appropriations. Research suggests there could be a higher risk of over-estimated caseloads if the forecast unit is located within the agency for which it forecasts. Additionally, because agency managers oversee the unit and make personnel decisions, forecast staff may feel pressure to incorporate management's concerns about possible budget shortfalls into their forecasts.

We found the forecasts were generally accurate in comparison to actual caseloads, with improved accuracy occurring between 2008 and 2011. Our analysis also showed the forecasts had no indication of overall statistical bias. We further determined the forecast methodology used was reasonable.

Although we found the forecasts had no indication of overall statistical bias, we found agency management had influenced the caseload estimate for the Aid to the Blind and Disabled program for the spring 2011 forecast. In

addition, we found the growth rate assumed for this forecast was not justified and also contributed to the inaccuracy of the forecast.

Actual caseloads have been lower than forecasted throughout most of the 2011-13 biennium for this program, resulting in the inaccurate spring 2011 forecast and a combined \$140.9 million (\$43 million General Fund) budget surplus through December 2012. Of the \$43 million General Fund surplus, we attribute approximately \$4 million to the influence of senior management, another \$19 million to an unjustified growth rate, and the remaining \$20 million to normal variance. The Legislature and Emergency Board reallocated this surplus to fund other needs.

Since the spring 2011 forecast, DHS and OHA have taken steps to address independence concerns. However, the current governance and oversight structures may not sufficiently protect the unit's independence. We developed two alternative governance and oversight options to further mitigate the risks to the forecast unit's independence and increase confidence in the caseload forecasts. The first alternative would create more independence by incorporating a policy oversight committee, but would retain the current administrative oversight. The second alternative would provide the most independence by placing the forecast unit under the administrative oversight of an external agency.

In addition, we found transparency weaknesses in the forecast unit's processes. For instance, advisory committee meeting minutes were not recorded and the unit did not post its methodologies or accuracy tracking reports on a public website.

We recommend DHS and OHA consider creating a policy oversight committee responsible for review and adoption of caseload forecast policies and procedures, and to help ensure forecaster independence. We also recommend the agencies increase the forecast unit's transparency by recording advisory committee meeting minutes, publishing an accuracy tracking report, and creating a public web page for this and other useful information about the unit.

We recommend the legislature consider the independence risks and the costs and benefits of alternative governance structures for the forecast unit.

Detailed recommendations appear on page 14 of the report.

Agency Response

The agency response is attached at the end of the report.

Background

The Office of Forecasting, Research, and Analysis (forecast unit) is a shared service of the Department of Human Services (DHS) and the Oregon Health Authority (OHA). The forecast unit forecasts monthly client caseloads for all eight major social service programs the two agencies manage (see Table 1). The two agencies use the caseload forecasts to develop program budgets, to monitor budget versus actual expenditures, and to make management decisions.

Table 1: DHS and OHA Caseload Forecasted Program Areas with Number of Individual Forecasts

Department of Human Services	Oregon Health Authority
Self-Sufficiency (6)	Medical Assistance Programs (14)
Child Welfare (4)	Addictions and Mental Health (8)
Vocational Rehabilitation (1)	Healthy KidsConnect (1)
Aging and People with Disabilities (13)	
Developmental Disabilities (13)	

Forecast unit staff are experienced, knowledgeable in statistical methods, and hold a range of advanced degrees in Economics, Demography, and Sociology. There are eight caseload forecast advisory committees, one for each major program, that review and advise on the forecasts.

The combined DHS and OHA 2011-13 Legislatively Adopted biennial budget was about \$20 billion, of which \$3.8 billion came from the General Fund. Although there is inherent uncertainty in forecasting caseloads, objective and impartial forecasts are important for developing accurate program budget estimates. Inaccurate caseload forecasts result in budget adjustments due to budget shortfalls or surpluses.

The Emergency Board approves budget adjustments, known as rebalances, between legislative sessions. This legislative committee is responsible for rebalances, allocating emergency funds, and other duties. In 18 of the last 42 meetings, the Emergency Board approved a DHS or OHA rebalance. Sometimes the two agencies requested additional funding along with a rebalance. Conversely, DHS returned approximately \$78.4 million back to the General Fund during a 2003 Emergency Board meeting. This return of funds helped to balance the statewide budget and was due in part to lower than estimated caseloads and program costs, and increased federal funding.

The accuracy of expenditure estimates may be even more critical with the expansion of Oregon Health Plan caseloads due to the Affordable Care Act. The Oregon Health Plan provides health care coverage to low-income Oregonians. Preliminary estimates suggest the Oregon Health Plan will add about 200,000 clients. According to a Kaiser Family Foundation report, this expansion will increase state expenditures by \$226 million over the next 10 years. However, there is increased uncertainty in forecasting the expanded number of clients due to a lack of historic caseload data. Not

having historical data means forecasts are inherently at a higher risk of being wrong and could lead to additional budget revisions in the future.

History of the forecast unit

Caseload forecasting for the state's social service programs has evolved over time. As early as the 1990's, DHS produced caseload forecasts using a decentralized process, with different divisions producing their own forecasts. The Department of Administrative Services (DAS) reviewed caseload forecasting in 1999 due to concerns about the process DHS used. Following the review, DHS developed a formal forecast methodology in conjunction with Willamette University. DHS also began efforts to create a centralized forecast unit. In 2005, DHS purchased new forecasting software and work began on the Integrated Client Services data warehouse. This data warehouse, which integrated DHS and OHA client data, allows forecasters to determine how many unique clients are served and how many clients intersect two or more program areas.

In late 2005, it became apparent that DHS had a \$172 million budget shortfall that was due in part to an under-estimated caseload forecast for the Aid to the Blind and Disabled program. Aid to the Blind and Disabled is a program within the Division of Medical Assistance Programs. In response to this large budget shortfall, the legislature formed a DHS taskforce in 2006 to understand the cause of the shortfall. At the same time, DAS began its second review of the forecast unit. As a result of these two reviews, the forecast unit improved its methodology and forecast reports, began producing monthly accuracy tracking reports, and increased staffing levels. In addition, Legislative Fiscal Office staff and DAS Budget Policy Analysts were included on the caseload forecast advisory committees.

Forecast uncertainty

Forecasts are inherently uncertain when numerous factors influence the outcome of the forecast. For example, social service caseloads are influenced by wages, employment levels, demographics, and prices of goods and services. Furthermore, many assumptions incorporated into a forecast, such as the future strength of the economy, are only estimates themselves. Time is also a factor. Predictions further out into the future are more uncertain.

The forecast unit produces forecasts for about 60 different social service programs administered by DHS and OHA. Forecast models incorporate a vast range of data from DHS, OHA, the Oregon Employment Department, the Office of Economic Analysis, Portland State University, and the U.S. Census Bureau. Generally, two different factors drive social service caseloads: economics and demographics. For example, the Supplemental Nutrition Assistance Program forecast is driven primarily by the state of the economy and rises rapidly during recessions. For this forecast, the forecaster models the number of new clients and the length of time they stay on the program by analyzing historic data. The model also incorporates employment levels and utilizes statistical techniques to

capture seasonal patterns. On the other end of the spectrum, the nursing facility forecast is driven primarily by demographics. For this forecast, the forecaster considers demographic projections. In addition to economic and demographic factors, state and federal policy can have a significant impact on program caseloads.

Forecasts are an important and critical policy tool, but users of these estimates should understand there is always a risk they will be wrong. For example, at the beginning of the most recent recession, many economists and professional forecasters inaccurately predicted both the onset and magnitude of the great recession.

Oregon's Office of Economic Analysis

The State of Oregon produces several other forecasts. The Office of Economic Analysis (OEA) produces many of these forecasts, including independent economic and revenue forecasts for statewide budgeting. OEA is also responsible for forecasting the state's population, and Department of Corrections and Oregon Youth Authority populations. The Department of Corrections was originally responsible for producing the corrections forecast, but this responsibility was moved to OEA to insulate the corrections forecast from potential bias. OEA receives input on its forecasts from advisory committees that include stakeholders, program staff, and economists.

Washington's Caseload Forecast Council

Created in 1997, Washington State's independent Caseload Forecast Council (Council) is responsible for producing caseload forecasts for all of the state's nondiscretionary programs, including K-12 education, corrections, and social service programs. Similar to Oregon, the agency responsible for administering Washington's social service programs produced caseload forecasts for these programs prior to 1997. The Washington State Legislature delegated forecasting duties to the Council in 1997 to address independence concerns and to remove politics and bias from statewide forecasting. Like Oregon, the Council's forecasters use advisory groups that include key stakeholders and program experts to provide input and advice for the forecasts. The forecast process also involves significant oversight from the Council, whose membership includes legislators and executives.

Audit Results

Forecast Methodology and Accuracy were Reasonable

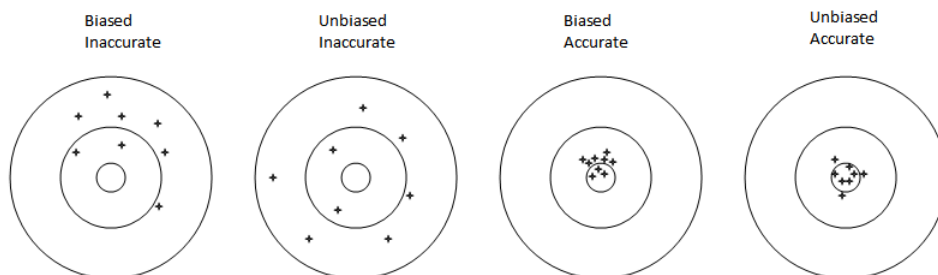
Forecast methodology was reasonable

We found the methodology the forecast unit uses is reasonable and forecasters follow sound practices. Also, economists we interviewed who are experienced in forecasting consider the forecast unit's models to be reasonable. The forecast unit's methodology consists of two main models: an *input-output* model and *time-series* analysis. The *input-output* model predicts client inflow, outflow, and transfers. *Time series* analysis uses historical and seasonal trends, and projects them into the future. In addition, the forecast unit sometimes uses expert consensus to estimate caseloads when new programs are developed and there is no historical data.

Forecasts are increasingly accurate with no indication of overall bias

A forecast's quality can be judged by two different measures: statistical bias and accuracy. Statistical bias is either systematic over or under-estimation of the forecast. Accuracy is a measure of how close the forecast matches actual results. An ideal forecast would be both unbiased and accurate (see figure 1).

Figure 1: Visual Bull's-eye Examples of Bias and Accuracy

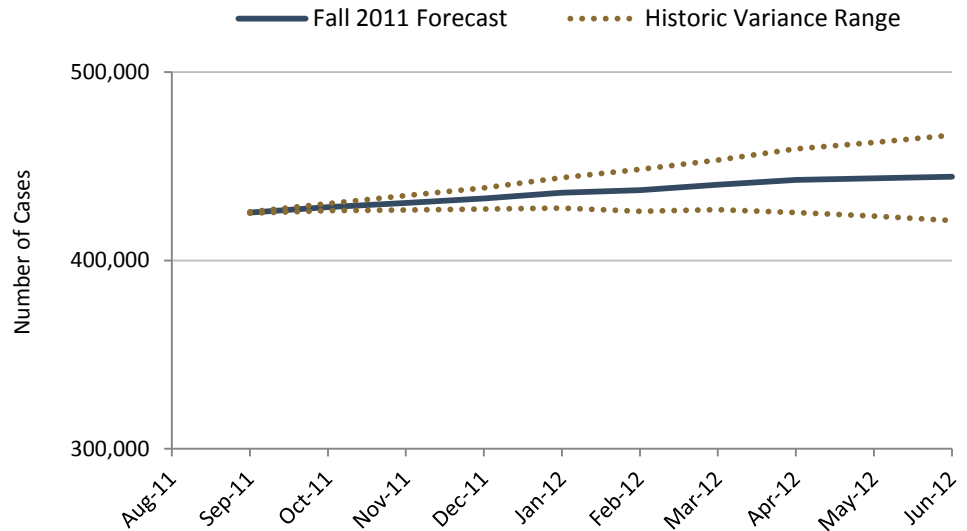


We analyzed accuracy and statistical bias by reviewing forecasts produced between 2008 and 2011. We compared actual caseloads against the first ten months of each forecast to identify variances. Forecast variance is the difference between the forecast and actual results. In some cases, we measured absolute variance, which is the difference between the forecast and actual caseloads regardless of whether the forecast was high or low.

We found forecasts to be generally accurate. For more than one-third of the forecasts, the difference between actual and forecasted caseloads was, on average, less than 1%. It is unreasonable to expect forecast variance to be zero because of all the uncertainty involved in forecasting. Additionally, we benchmarked a limited number of Oregon's forecasts to those produced by Washington's Caseload Forecast Council. Both states had similar performance, although Washington's accuracy was marginally higher.

As expected, we found accuracy was highest in the near-term and decreased for projections further out into the future. In figure 2, we overlaid historic variances for the Supplemental Nutrition Assistance Program onto the fall 2011 forecast. The historic variance range represents the average variance exhibited in forecasts since 2008. We believe future caseload variances are likely to fall within this range. Forecasts made further out in the future have the largest historic variances.

Figure 2: Fall 2011 Supplemental Nutrition Assistance Program Forecast with Historic Variance Range



Note: Non-zero origin used to emphasize differences.

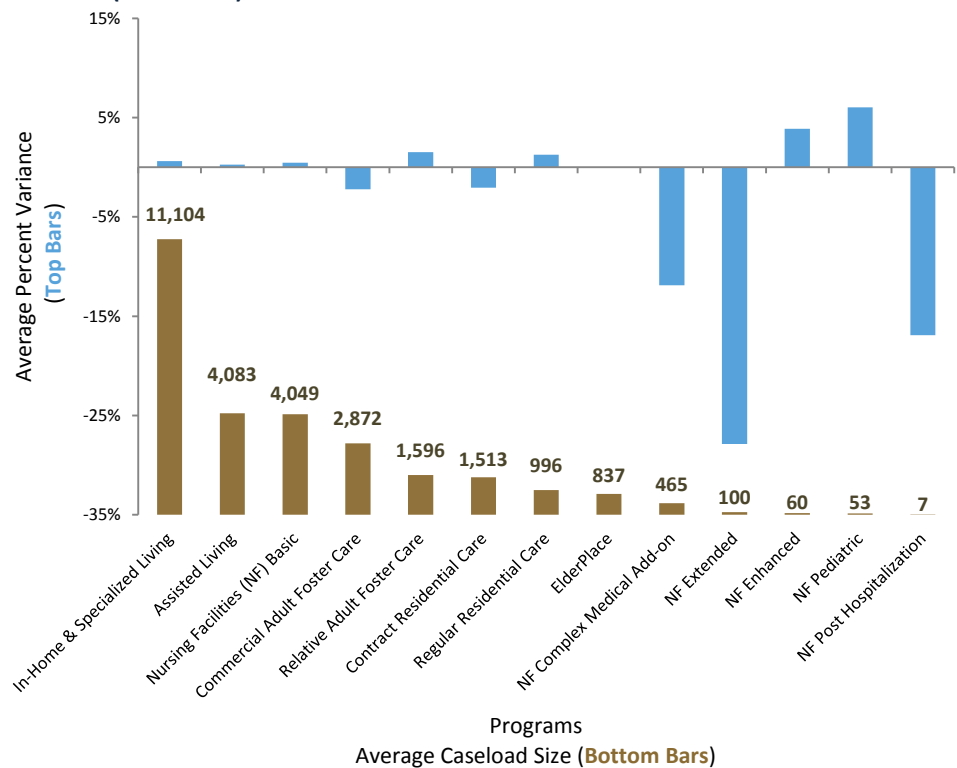
We found statistically significant improvement in forecast accuracy between 2008 and 2011. For example, as shown in Table 2, Self Sufficiency and Vocational Rehabilitation programs had a combined 4.5% variance between forecasted and actual caseloads in 2008 that improved to about a 1% variance in 2011. While there was no one specific cause for the improvement in forecast accuracy, recent stability in the economy is one likely factor. Another contributing factor could be improved forecast models. For example, two of the Self Sufficiency forecasts recently incorporated employment data. Another possible cause is increased use of the Integrated Client Services data warehouse. This data warehouse is an extremely useful tool that allows forecasters to track clients across all DHS and OHA programs. Regardless of the cause, increased forecast accuracy is a good indication of the forecast unit's performance.

Table 2: Forecast Variance between 2008 and 2011

Program	2008 Forecasts	2011 Forecasts
Self Sufficiency and Vocational Rehabilitation	4.5%	1.0%
Aging and People with Disabilities	1.3%	1.1%
Medical Assistance and Healthy KidsConnect	3.4%	2.5%
Overall	3.1%	1.5%

We also did not identify any patterns of overall bias in the forecasts produced between 2008 and 2011. Overall, recent forecasts were not systematically over-estimated. Although some forecasts were over-estimated, others were under-estimated and several forecasts were very accurate. As expected, small programs tended to have larger variances, while large programs tended to have smaller variances. For example, Figure 3 shows Aging and People with Disabilities programs sorted by average caseload size (bottom bars) and each program's average variance (top bars). Even though there were large variances for small programs with few cases, the programs with the larger caseloads were much more accurate, resulting in a small, -0.1% overall variance.

Figure 3: Caseload Size and Forecast Variance of Aging and People with Disabilities Forecasts (2009-2011)



Independence Issues

Accurate caseload forecasts are important for budgeting

The DHS and OHA budget units use caseload forecasts to estimate expenditures. It is important for caseload forecasts to be accurate. If estimated expenditures are over-stated, the legislature may initially approve a larger budget than necessary to meet actual demand. Because some of these programs are so large, a caseload forecast that is just 1% high could cause the program to be over budget by as much as \$17.8 million. This would divert dollars from other state needs.

Yet, an under-estimated forecast may require a new appropriation from state General Fund reserves. Agency requests to the legislature for new appropriations can be uncomfortable and seen by management as something to avoid. For example, after DHS had a \$172 million budget shortfall in 2005, the legislature formed a taskforce to evaluate the agency's effectiveness, including the accuracy of its caseload forecasts.

Agencies may have an incentive to create a cushion in their budgets by over-estimating caseload forecasts. A budget surplus allows an agency to cover budget shortfalls without requesting new appropriations. Research suggests there could be a higher risk of over-estimated forecasts if the forecast unit is located within the agency for which it forecasts. Additionally, because agency managers oversee the unit and make personnel decisions, forecast staff may feel pressure to incorporate management's concerns about possible budget shortfalls into their forecasts.

Even if there is no real threat to forecaster objectivity, there can be a perception by the State Legislature or the public of biased forecasts.

Lack of independence contributed to an inaccurate program forecast

We found the forecast unit's lack of organizational independence contributed to an inaccurate caseload forecast for one program. In spring 2011, the caseload forecast for the Aid to the Blind and Disabled program was changed as a result of senior management's influence. In addition, we found the growth rate assumed in the forecast was not justified and contributed to the inaccurate forecast.

The forecast unit produced the spring 2011 forecast as part of the 2011-2013 biennium budget development process. The Aid to the Blind and Disabled portion of the forecast was produced after the forecaster obtained input and advice from the program's caseload forecast advisory committee.

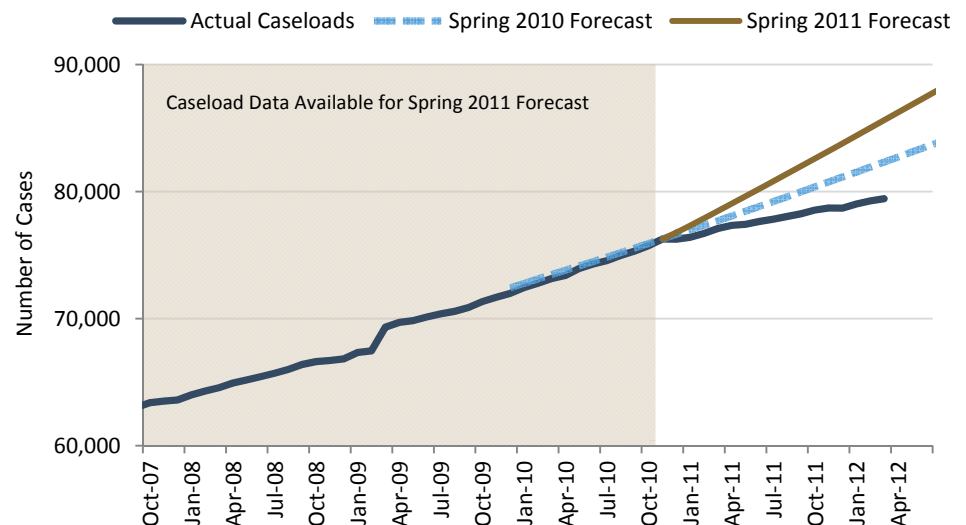
After the forecast number was produced, the DHS director at the time met with the forecaster, the forecast unit administrator, and the division administrator. The division administrator told us the former DHS director

was concerned that if the Aid to the Blind and Disabled forecast was underestimated, it would result in a severe budget shortfall. The director was concerned because in 2005 the program faced a large budget shortfall caused in part by an under-estimated forecast. Based on these concerns, the DHS Director asked the forecaster to look at the Aid to the Blind and Disabled forecast again and consider incorporating economic factors. Following the meeting, the forecast was increased by 702 cases.

We also found that carrying forward the growth rate from the fall 2010 forecast to the spring 2011 forecast, was not justified. This assumed growth rate was well above the historic range and added an additional 3,200 cases to the spring 2011 forecast. The forecast unit's fall 2010 report attributed the upward shifts to the aging population. Historically, the program's caseload growth was relatively stable and predictable. The program also did not appear to be significantly influenced by economic factors during the recession.

The actual caseload history available at the time of the spring 2011 forecast showed the previous spring 2010 forecast was precisely tracking the actual caseloads and had almost no variance. Figure 4 shows the actual caseload history line (bottom), the spring 2010 forecast line (middle), and the spring 2011 forecast line (top). Finalized historical caseload data through at least October 2010 would have been available at the time the spring 2011 forecast was increased (shaded region).

Figure 4: Aid to the Blind and Disabled Caseload and Forecast History



Note: Non-zero origin used to emphasize differences.

Table 3 shows the history of the semiannual caseload forecasts produced for the 2011-13 biennium budget process for the Aid to the Blind and Disabled program. The first three forecasts were preliminary estimates for the 2011-13 budget. The first spring 2011 forecast was the original forecast that resulted from the caseload forecast advisory committee process. The revised spring 2011 forecast was used for developing the 2011-13 OHA

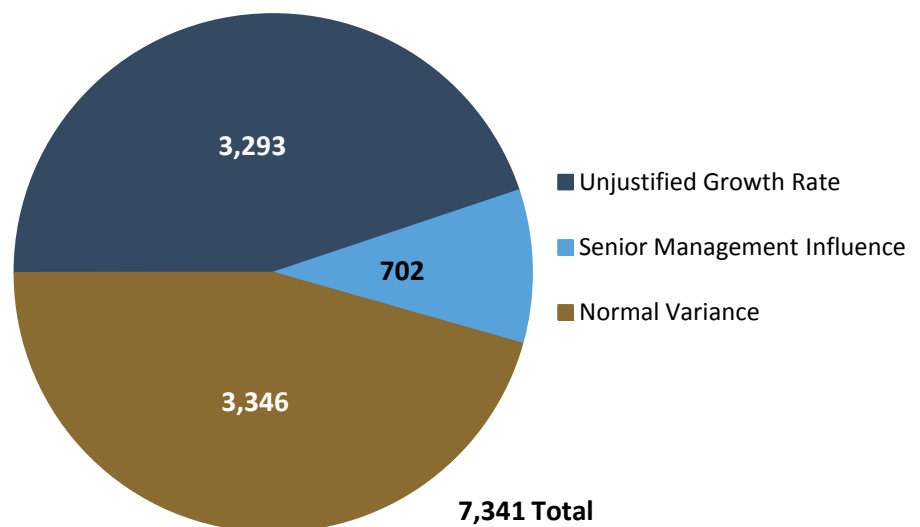
budget and became part of the Legislatively Adopted Budget. The last three forecasts were developed as part of the rebalance process for the 2011-13 budget.

Table 3: History of Caseload Forecasts Used for the Aid to the Blind and Disabled 2011-13 Biennium Budget Process

Forecast Use	Date	ABAD Caseload Forecast	Change from Prior Forecast
Preliminary Estimate	Fall 2009	83,771	
Preliminary Estimate	Spring 2010	83,771	0
Preliminary Estimate	Fall 2010	87,008	+ 3,237
Develop 2011-13 OHA Budget	Spring 2011 (Original)	87,064	+ 56
Develop 2011-13 OHA Budget	Spring 2011 (Revised)	87,766	+702
Rebalance 2011-13 Budget	Fall 2011	82,593	- 5,173
Rebalance 2011-13 Budget	Spring 2012	80,524	- 2,069
Rebalance 2011-13 Budget	Fall 2012	80,425	- 99

The legislatively adopted 2011-13 budget for the Division of Medical Assistance Programs was \$6.6 billion. Of that, \$1.78 billion (\$549.9 million General Fund) was designated for the Aid to the Blind and Disabled caseload. However, the actual caseloads have been lower than forecasted throughout most of the 2011-13 biennium for this program, resulting in the inaccurate spring 2011 forecast and a combined \$140.9 million (\$43 million General Fund) budget surplus through December 2012. The calculated budget surplus is based on the fall 2012 caseload forecast. The fall 2012 caseload forecast was used for rebalancing the 2011-13 budget and incorporated available actual caseloads. The difference between the fall 2012 forecast and the spring 2011 forecast is 7,341 fewer cases. Of this variance in caseloads, we attribute 702 to the influence of senior management, 3,293 to the unjustified growth rate, and 3,346 to normal variance in caseload growth (see Figure 5).

Figure 5: Breakdown of the Difference between the Fall 2012 and Spring 2011 Forecast for the Aid to the Blind and Disabled Caseload by Reason



We attribute approximately \$4 million of the \$43 million General Fund surplus to the influence of senior management. We attribute about \$19 million of the General Fund surplus to the unjustified growth rate. The remaining \$20 million of the General Fund surplus is due to normal variance in caseload growth. The Legislature and Emergency Board reallocated this surplus to help cover the cost of under-estimated caseloads for other social service programs and federally required updates to the Medicaid Management Information System.

Concerns have been expressed about the forecast unit's independence. Following each forecast cycle, the forecast unit surveys all advisory committee members to assess a variety of customer service metrics as well as members' perceptions on the unit's objectivity, transparency, and independence. During one survey, one respondent noted that the unit "could be required by politics and the executives to whom they report to modify forecasts, and therefore, not be allowed to exercise the independence they want and need to generate credible forecasts."

Likewise, one state senator and one state representative had questions about the caseload forecasts and expressed concerns to us about the forecast unit's independence and the potential for managers to influence the forecasts. They noted biased forecasts could result in over-estimated caseloads and budget surpluses that could have been used elsewhere within the State's budget. Furthermore, external stakeholders expressed similar concerns.

Increasing Independence

Having the forecast unit placed within DHS and OHA means the two agencies are responsible for caseload forecast accuracy. The agencies are also accountable for forecast estimates that cause programs to be either under budget or over budget. One incentive for the agency to create accurate forecasts is legislative oversight and the consequences that result when inaccuracies create budget problems.

There are advantages to having the forecast unit within DHS and OHA. For instance, forecasters have direct access and close proximity to agency staff, many of whom have extensive, detailed knowledge of state and federal program policy changes that influence caseloads. Having this knowledge benefits forecast accuracy. Another advantage is forecasters having direct access to agency data, which includes historic caseload data. Because the greatest technical knowledge and expertise lies within the agency, forecasters placed within the agency will have quicker access to information. Last, notwithstanding the independence concerns around the spring 2011 forecast, the DHS or OHA director may be able to provide input that could improve the forecast. The forecast unit also provides an array of supplemental research, analysis, and consulting services that would still be needed by DHS and OHA. Thus, moving the forecast unit outside of the agency is not without cost or consequence to the agencies' programs.

Since the spring 2011 forecast, DHS and OHA have taken further steps to address independence concerns. They developed two formal governing documents for the forecast unit, a service level agreement and a committee charter for the eight-caseload advisory committees. The service level agreement includes the unit's purpose and responsibilities, and designates guidance and oversight of the unit to the DHS and OHA Joint Operations Steering Committee, which is the main oversight body for both agencies. The forecast advisory committee charter includes the purpose, membership, roles, and responsibilities of the committees. According to agency management, the advisory committee structure ensures the objective development, review, and approval of caseload forecasts.

However, the current governance and oversight structures may not adequately protect the unit's independence, in actuality or appearance. As a shared service of DHS and OHA, the forecast unit administrator reports to both DHS's Chief Financial Officer and OHAs Budget Director. Thus, DHS and OHA management oversee forecast unit personnel decisions, such as hiring, promoting, and disciplining forecast staff.

Research found that Washington's forecast accuracy increased after forecasts were produced independently. Based on this research and after reviewing other government forecast units, we developed two alternative governance and oversight options. These governance options would further mitigate the risks to the forecast unit's independence and increase confidence in the caseload forecasts.

More independence with external advisory committee

The first alternative would create more independence for the forecast unit, but would retain the current administrative oversight. A committee that includes external experts, such as economists or other experts from the public, academic, non-profit, or private sectors, would be responsible for policy oversight of the forecast unit. This oversight committee would review and adopt caseload forecast policies and procedures. It would help ensure the independence of the forecast unit by providing an avenue for forecasters to voice their concerns beyond agency management.

The oversight committee would be distinct from the existing eight advisory committees, which would continue to provide input on risks, assumptions, and methodology. The advisory committees would also continue to provide advice on the caseload forecasts. However, these advisory committees would be strengthened by adding additional external representation from the public, academic, non-profit, and/or private sectors.

Most independence with forecasters outside the agency

The second alternative would provide the most independence for the forecast unit. It would include the policy oversight and advisory committee functions as described in the first alternative, but would also place the forecast unit under the administrative oversight of an external agency. This structure would help protect forecasters from real or perceived pressure by DHS or OHA management to change a forecast. The unit would remain near the program staff to ensure good communications.

We found several examples of other forecast units with external administrative oversight, which appear to work quite well. For example, the unit that forecasts Oregon's corrections population is located within the Office of Economic Analysis. This office has administrative oversight by the Department of Administrative Services. Washington State's Caseload Forecast Council is also an independent agency that produces all nondiscretionary caseload forecasts for the state.

More transparency needed

Increased transparency is important for ensuring accountability. Lack of transparency can lead to distrust by lawmakers and the public. During our review, we found transparency weaknesses in the forecast unit's caseload forecast process. The forecast unit has posted its semiannual caseload forecast and other research reports on two shared agency web pages. However, there was no dedicated web page for the forecast unit. Also, the forecast unit has not posted to the web caseload forecast methodologies, advisory committee membership, or forecast accuracy tracking reports.

Some state legislators do not understand the forecast unit or its processes very well. For example, one state legislator was not aware of the general methodology the forecast unit uses. Legislators we talked to would also like to know the forecast unit's accuracy over time. Additionally, increased transparency of the forecast process, methodologies, and assumptions

could instill greater confidence in the forecasts. Ultimately, additional transparency could allow the Oregon Legislature to make more informed budget allocation decisions.

The eight caseload forecast advisory committees do not record meeting minutes. For example, there are no meeting minutes that support the significant increase in caseloads between the spring and fall 2010 forecasts. Recording meeting minutes that at a minimum include the key information or issues discussed and the committee's advice on the forecasts would increase transparency.

The forecast unit measures and reviews its forecast quality by comparing actual caseloads to forecasts and provides monthly tracking reports to advisory committee members, agency staff, and other interested stakeholders. However, the unit does not make this information available on the agencies' website. In addition to monthly reports, the unit could publish annual or semiannual accuracy tracking reports that include both accuracy and statistical bias information to provide key information to legislators and other forecast report users. These reports could also be used internally to further improve the quality of the unit's forecasts.

We also reviewed the publicly available semiannual caseload forecast reports for transparency. We found that the reports could provide more detail on risks, assumptions, uncertainties, and how these factors could affect caseload estimates. In addition, while these reports have improved in recent years, additional details and context would enhance the reports for the average reader.

During our review, management indicated a commitment to increasing transparency and noted that the forecast unit is making plans to develop a public website. In addition to the semiannual forecast reports, the forecast unit plans on including the following on the website: governance documents, performance measures, advisory committee membership, forecast methodology, data sources, monthly accuracy tracking reports, and other work products.

Recommendations

To improve the independence, oversight, and transparency of the forecast unit, we recommend the Department of Human Services and the Oregon Health Authority:

- Consider creating a policy oversight committee responsible for review and adoption of caseload forecast policies and procedures, and to help ensure forecaster independence.
- Continue using the eight caseload forecast advisory committees as the arena to debate forecast risks, assumptions, and methodology, and to advise the forecasters on the caseload forecast numbers.
- Consider adding additional external representation to the eight advisory committees from the public, academic, non-profit, and/or private sectors.
- Record meeting minutes of the eight advisory committees that at a minimum include the key information or issues discussed and the advisory committee's advice on the forecast numbers.
- Regularly evaluate the forecasts and publish an annual accuracy tracking report. Include in the tracking report an analysis of both the accuracy and statistical bias of the forecasts. Use the results to identify improvements in assumptions and methodologies.
- Continue efforts to create a public web page dedicated to the forecast unit, and post its methodologies, advisory committee membership, advisory committee meeting minutes, forecast accuracy tracking reports, and the semiannual caseload forecast reports.
- Include in the published semiannual caseload forecast reports additional detail on risks, assumptions, uncertainties, and how these factors could affect caseload estimates.

We recommend the Legislature consider the independence risks and the costs and benefits of alternative governance structures for the forecast unit.

Objectives, Scope and Methodology

We conducted this audit at the request of State Senator Richard Devlin, OHA Director Dr. Bruce Goldberg, and the State's Chief Operating Officer Michael Jordan. The objective of our audit was to determine if adequate independence, oversight, and transparency exists for the forecast unit to ensure objective and reasonable caseload and expenditure forecasts.

The scope of our audit included (1) conducting a general review of forecast methodology, (2) analyzing the forecasts produced by the forecast unit to determine if they are accurate and unbiased, and (3) reviewing the independence and transparency of the forecast unit.

To answer our audit objectives, we reviewed Oregon laws and rules, and Department of Human Services and Oregon Health Authority charters, policies, and budget documentation. We interviewed forecast staff, budget staff, policy staff, program staff, and senior management to gain an understanding of the social service forecasting process and related internal controls at DHS and OHA. We also interviewed experts from Louisiana, Minnesota, New York, Oregon, South Carolina, Washington State, and Washington, D.C. We reviewed academic articles on forecasting practices, and received assistance from the Congressional Budget Office and Washington State's Caseload Forecast Council. We also considered information received through our Government Waste Hotline.

We obtained two sets of data from the forecast unit. The first set was data from eight consecutive caseload forecasts from 2008 through 2011. The second set contained actual caseload counts from 2001 through 2012. The data encompassed approximately 56 different social service programs. We assessed the data for reliability and sufficiency. We verified that we received all data we requested. We also tested for duplicate and missing data. We compared a sample of data to files maintained by the agency and reviewed the Integrated Client Services data warehouse. As a result of these procedures, we determined the data to be sufficiently reliable for our audit purposes.

We obtained forecasts for the Addictions and Mental Health, Aging and People with Disabilities, Developmental Disabilities, Medical Assistance, and Self Sufficiency programs. We analyzed all forecasts, but limited our interpretation of results for the Developmental Disabilities and Addictions and Mental Health programs. We limited our review of the Developmental Disability and Addictions and Mental Health forecasts because they were implemented after spring 2008 and detailed Addictions and Mental Health forecasts are not used for budget purposes. We also did not analyze Child Welfare forecasts because of limitations related to the implementation of the OR-Kids data system and the development of new forecast methodologies.

We analyzed the accuracy of each forecast from spring 2008 to fall 2011 by comparing forecasts to actual caseloads for the ten months that follow each

forecast. We also looked for systemic bias using various statistical methods, including weighting averages on caseloads and dollars. Weighted averages consider the size of each program when calculating the average, so larger programs contribute more to the average than smaller programs. We followed up with agency staff when we observed suspected biases.

We also reviewed management controls within the DHS and OHA budget units, and found adequate management controls were in place over the calculation of cost per case estimates and other budget calculations.

We conducted this performance audit in accordance with generally accepted government auditing standards. Those standards require that we plan and perform the audit to obtain sufficient, appropriate evidence to provide a reasonable basis for our findings and conclusions based on our audit objectives. We believe that the evidence obtained provides a reasonable basis for our findings and conclusions based on our audit objectives.

February 5, 2013

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RE: Department of Human Services and Oregon Health Authority Response to *Health and Human Service Caseload Forecasting: Ways to Increase Confidence* Draft Audit Report

Dear Mr. Blackmer:

Thank you for the opportunity to respond to the draft audit entitled, *Health and Human Service Caseload Forecasting: Ways to Increase Confidence*. The Department of Human Services (DHS) and Oregon Health Authority (OHA) generally agree with the recommendations made in the draft report. Per your request, below is the agencies' combined response to the DHS and OHA related recommendations. We have also provided a discussion section in this letter to provide additional information and context to our response.

Recommendation:

To improve the independence, oversight, and transparency of the forecast unit, we recommend the Department of Human Services and the Oregon Health Authority:

- Consider creating a policy oversight committee responsible for review and adoption of caseload forecast policies and procedures, and to help ensure forecaster independence.
- Continue using the eight caseload forecast advisory committees as the arena to debate forecast risks, assumptions and methodology, and to advise the forecasters on the caseload forecast numbers.
- Consider adding additional external representation to the eight advisory committees from the public, academic, non-profit, and/or private sectors.
- Record meeting minutes of the eight advisory committees that at a minimum include the key information or issues discussed and the advisory committee's advice on the forecast numbers.

- Regularly evaluate the forecasts and publish an annual accuracy tracking report. Include in the tracking report an analysis of both the accuracy and statistical bias of the forecasts. Use the results to identify improvements in assumptions and methodologies.
- Continue efforts to create a public web page dedicated to the forecast unit, and post its methodologies, advisory committee membership, advisory committee meeting minutes, forecast accuracy tracking reports, and the semiannual caseload forecast reports.
- Include in the published semiannual caseload forecast reports additional detail on risks, assumptions, uncertainties, and how these factors could affect caseload estimates.

Discussion:

DHS and OHA agree that accurate caseload forecasts are critical for agency budgeting and legislative allocation of limited state funds. Since both agencies are required to report to the Emergency Board and would prefer to avoid unnecessary rebalances, both agencies have a strong incentive to support forecasts which are neither too high, nor too low.

As documented in the report, the DHS and OHA Office of Forecasting, Research and Analysis (OFRA) has knowledgeable staff, uses reasonable methodologies, and produces forecasts that are generally accurate, have improved over time, and have no indication of overall bias. It was also determined that OFRA's forecasts are only marginally less accurate than those produced by Washington State's independent Caseload Forecasting Council. OFRA performs an array of supplemental research, analysis, and consulting services for DHS and OHA that would still be needed if the unit was moved out of the agencies.

The report also notes that forecasts are inherently uncertain for a variety of reasons. However, DHS and OHA believe that federal and state policy and procedures have the largest impact, with economic and demographic impacts secondary. While the forecasting unit strives for accuracy, subsequent adjustments and corrections are to be expected. The realistic expectation of inaccuracy is one of the primary reasons the forecast is updated twice each year and incorporated into the budget adjustment requests submitted to the Emergency Board.

The report describes how the forecasted 2011-13 caseload for Aid to the Blind and Disabled program (ABAD) varied over the course of seven forecast cycles. DHS and OHA generally agree with the basic facts as presented, but not with the view expressed in

the report that the error was due to a lack of independence or that more than half of the error was not justified by the information available at that time. The majority of the caseload increase which ultimately proved to be wrong was discussed and approved in the fall 2010 forecast by the Medical Assistance Program Caseload Advisory Committee, which included staff from both the Department of Administrative Services (DAS), Budget and Management Division (BAM) and the Legislative Fiscal Office (LFO). While not noted in the Fall 2010 DHS and OHA Caseload Forecast report, one of the factors that contributed to the inaccuracy was the fact that Social Security had experienced a steep increase in disability applications since the Great Recession began. This had been widely reported in the national press, so forecasting unit staff considered it reasonable to anticipate some increase to the rate of growth for the ABAD caseload.

DAS Budget Policy Analysts and LFO staff have participated in the caseload forecasting process for many years. In addition, since DHS and OHA split into separate agencies in July 2011, formalized governance processes have been created to oversee all shared services units, including forecasting. As mentioned in the report, the forecast unit administrator reports to the DHS Chief Financial Officer and the OHA Budget Director, and the unit itself operates under authority of the DHS and OHA Joint Operations Steering Committee (JOSC). JOSC has authorized two governance documents that pertain to the forecasting unit: a Service Level Agreement which lays out the unit's key responsibilities and deliverables, and a Caseload Forecast Advisory Committee Charter which specifies the authority, roles, and responsibilities of the committees, their members, and the forecast unit.

The forecasting unit has also documented key processes and established performance metrics. Documentation includes monthly forecast accuracy reports, methodology(s) used to create the forecast for each caseload area, key process flow charts, an annual calendar for scheduled work, and a unit performance scorecard.

DHS and OHA strive to be transparent. To that end, an internet website has been created that is expected to go live within the next two weeks. It will be easily accessible from both the DHS and OHA home sites. The new website will include work products (e.g. caseload forecasts and monthly variance reports), administrative documents (e.g. materials on governance and methodology, names of advisory committee members, etc.), and links to other forecasting units. DHS and OHA have been routinely posting forecasts online and the bi-annual forecasts going back to 2006 are currently available.

Agencies Response:

The agencies generally agree with the recommendations made directly to DHS and OHA within the report.

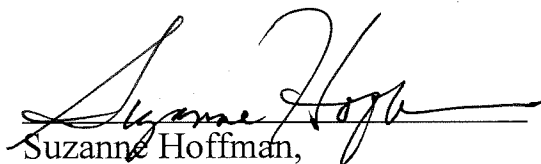
- DHS and OHA will ask for assistance from the DAS Office of the Chief Operating Officer (COO) and LFO to establish a Forecasting Policy Oversight Committee. This effort will include establishing the committee's charter, including purpose, general membership, and meeting frequency. We plan to have these steps completed by April 2013. It is also our hope that DHS, OHA, DAS and LFO will also work together to recruit members for this and other individual caseload committees. We will target the first committee meeting to occur prior to October 2013. One of the decisions for the committee, once its initial tasks are complete, is whether it is an ongoing committee or one that only meets infrequently to make any necessary reassessments.
- DHS and OHA will continue to utilize the eight forecast advisory committees as described in the audit report.
- As indicated above, DHS and OHA will ask for assistance from DAS COO and LFO to identify appropriate external members in time for the fall 2013 forecast advisory committee meetings in October 2013. In addition, as staffing levels allow, the forecasting unit will continue to periodically survey external stakeholders who are not on the committees to get their input on the factors driving caseload changes.
- Starting with the spring 2013 forecast advisory committee meetings in February 2013, the forecast unit will ensure that written records of decisions and action items are kept for each forecast advisory committee meeting.
- DHS and OHA understand the importance of tracking and reporting forecast accuracy and using that information to identify possible improvements. Starting February 2013, the forecasting unit will begin posting the existing monthly variance reports on the internet. Starting April 2013, the forecasting unit will begin posting the unit's Quarterly Performance Scorecard, which includes measures of forecast accuracy, to the internet. By October 2013, the forecasting unit will publish an annual accuracy tracking report.
- DHS and OHA will create and maintain a website dedicated to the forecasting unit. Work on development of a website began before the audit and an initial website has been created. Links to the final site will be added to both the DHS and OHA home sites by mid-February 2013. The new website includes forecast unit work products (e.g. caseload forecasts and monthly variance reports), administrative

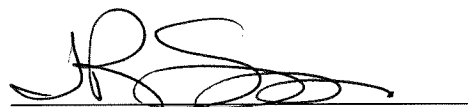
documents (e.g. materials on governance and methodology, names of advisory committee members, etc.), and links to other forecasting units.

- DHS and OHA understand the importance of explaining the uncertainties and assumptions which underlie the forecasts and the factors that pose the greatest risk to forecast accuracy. Starting with the spring 2013 forecast publications, forecast unit staff will work to strengthen how these subjects are covered within the semi-annual forecast publications.

Thank you again for the opportunity to address the recommendations included in the draft audit report. Please feel free to contact Eric Moore, DHS Chief Financial Officer; William Coulombe, OHA Budget Director or Dave Lyda, Chief Audit Officer for DHS and OHA if you have any questions regarding this response.

Sincerely,


Suzanne Hoffman,
OHA Chief Operating Officer


Jim Scherzinger,
DHS Chief Operating Officer

cc: Bruce Goldberg, OHA Director
Erinn Kelley-Siel, DHS Director
Eric Moore, DHS Chief Financial Officer
William Coulombe, OHA Budget Director
Betsy Jensvold, OFRA Administrator, DHS and OHA
Dave Lyda, Chief Audit Officer, DHS and OHA

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The courtesies and cooperation extended by officials and employees of the Department of Human Services and Oregon Health Authority during the course of this audit were commendable and sincerely appreciated.